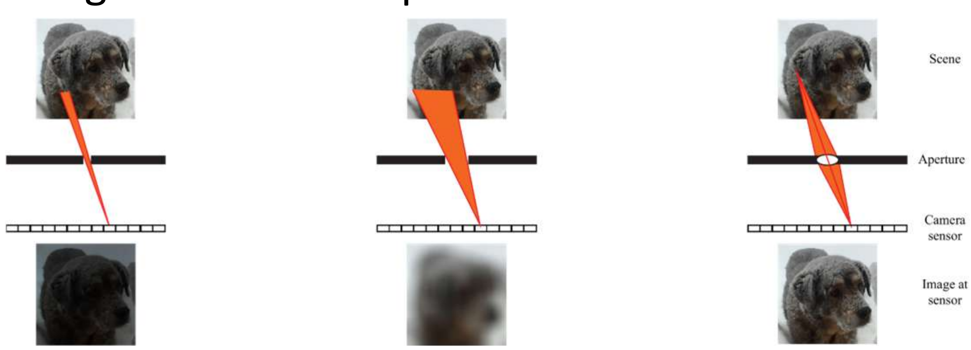
# Agenda: Computer Vision Camera is a Matrix

1

# What is a camera

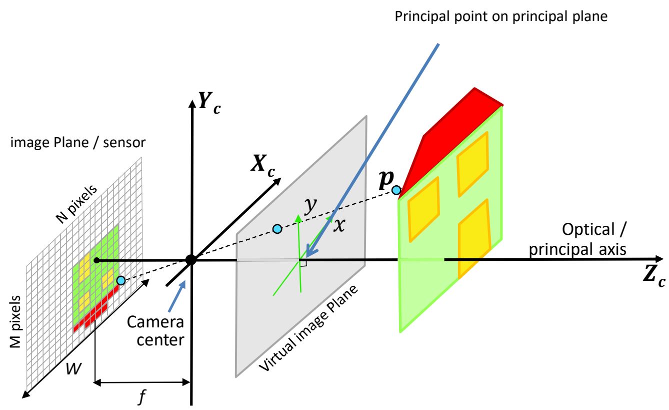
- Is the pin-hole camera model is OK?
- Image formation requires a lens




2

# Agenda: Camera as a Matrix

- Computer vision cameras are modelled as pinhole camera model
- A camera is a matrix
  - What?
  - $x_{3\times1} = PX$
  - The equation is a homogeneous equation



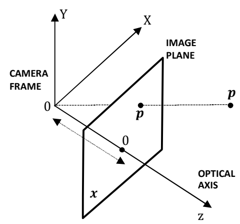

3

# Homogenous

- A homogeneous system is equivalent to a matrix equation of the form $Ax = 0$
  - If $x$ is a solution, then so $kx$
- Our camera matrix equation $x = PX$ should actually be understood in the same way
  - $x$ and $kx$ both are deemed to satisfy the equation
  - Conversely $P$ is valid “upto a constant”
- The representation $x$ of a geometric object is homogeneous if $\boldsymbol{x}$ and $k\boldsymbol{x}$ represent the same object ($k \neq 0$)

4

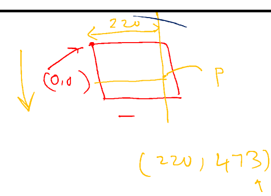
# Terminology



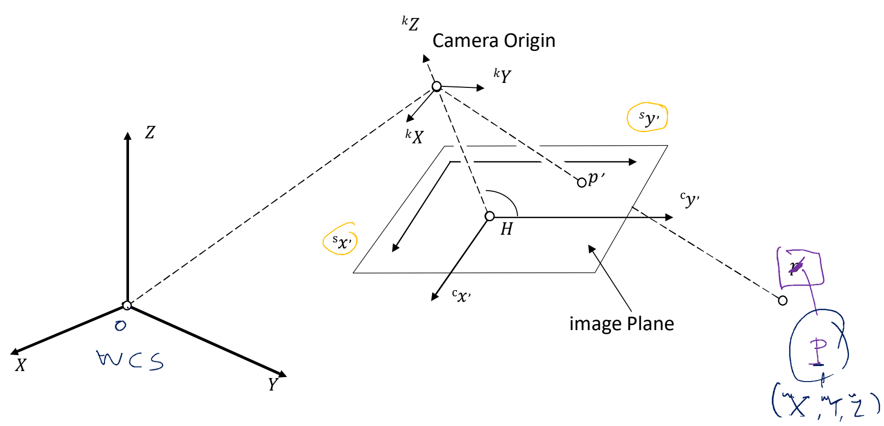

- **O** = camera center (pinhole), also treated as origin of the coordinate system for the object ("camera coordinate system" or "camera frame")
- $\boldsymbol{P} = (X, Y, Z)$ = object point (3D vector)
- $\boldsymbol{p} = (x, y, z = f)$ image of $\boldsymbol{P}$ on the image plane $\pi$,
- $\boldsymbol{p}$ is obtained by the intersection of line $OP$ with $\pi$
- principal axis (optical axis) set up as $Z$ axis is normal to $\pi$
- $\pi$ is parallel to $XOY$ plane (principal plane)
- $\boldsymbol{o}$ = principal point the point where the vector normal to π and passing through $\boldsymbol{O}$ intersects $\pi$



# Camera as a Matrix



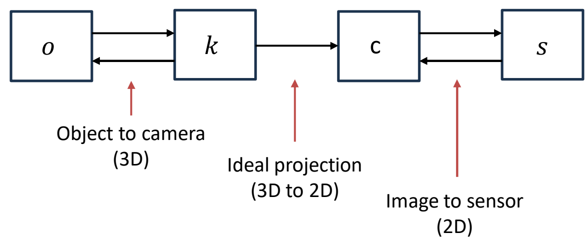

Four elements



# World to Sensor

$s = P o$

1. World coordinate system
2. Camera Coordinate System
3. Image (planar) coordinate system
4. Sensor coordinate system




7

# Cartesian Coordinate Systems

1. World coordinate system

Euclidean

$(X,\ Y,\ Z)^T$

2. Camera Coordinate System

$({}^kX,\ {}^kY,\ {}^kZ)^T$

3. Image (planar) coordinate system

$({}^cx,\ {}^cy)^T$

4. Sensor coordinate system $({}^sx,\ {}^sy)^T$

Until this course!

8

# Homogeneous Coordinate Systems

1. World coordinate system $[X,\ Y,\ Z,\ 1]^T$
2. Camera Coordinate System $\left[{}^kX,\ {}^kY,\ {}^kZ,\ 1\right]^T$
3. Image (planar) coordinate system $[{}^cx, {}^cy, w]^T$
4. Sensor coordinate system $({}^sx, {}^sy, {}^sw)^T$

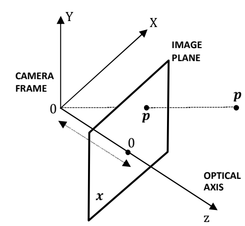
Until this course!

Homogeneous

9

# Camera Matrix

- To understand this sequence start with projection with what we already know
- The relationship between $(x, y)$ and $(X, Y, Z)$ has been linearized using homogeneous coordinates on both the left and right hand sides




$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

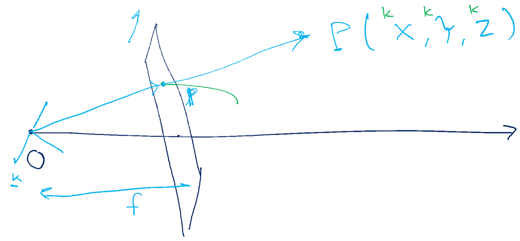
$$x = \frac{x'}{z'}, \quad y = \frac{y'}{z'}$$

10

$c_x = fX$
$c_y = fY$
$w = Z$

$$\boxed{c_x = \frac{fX}{Z}}$$

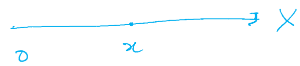
o k c s




$$x\begin{pmatrix} c_x \\ c_y \\ w \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$\vec{x} = \begin{bmatrix} f & f & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} X \\ Y \\ Z \\ 1 \end{matrix}$$

$\vec{Op} = k\vec{OP}$
$x = kX$ ÷ 1
$y = kY$ 2
$z = kZ$
$z = f \Rightarrow k = \frac{f}{Z}$




$x = \frac{f}{Z} \cdot X$ ✓ $= \mathrm{diag}(f).[I\ \vec{0}]$

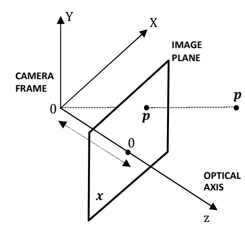
$y = \frac{f}{Z} \cdot Y$

$$\begin{pmatrix} c_x \\ c_y \\ w \end{pmatrix} = \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

11

# Camera Matrix

- To understand this sequence start with projection with what we already know
- The relationship between $(x, y)$ and $(X, Y, Z)$ has been linearized using homogeneous coordinates on both the left and right hand sides




$p(x, y)$

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$x = \frac{x'}{z'}, \quad y = \frac{y'}{z'}$$

$$\boxed{o} \rightleftarrows \boxed{k} \rightarrow \boxed{c} \rightleftarrows \boxed{s}$$

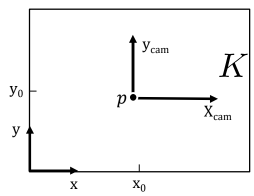
Where are we in this picture?

$x_{3\times1} = P_{3\times4} X_{cam}$ where $P = \mathrm{diag}(f, f, 1)[I_{3\times3} | 0]$

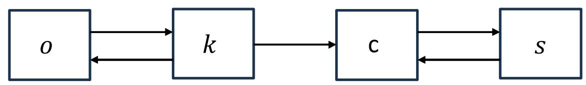
12

# Camera Matrix

- We know
  $x_{3\times1} = P_{3\times4}X_{cam}$ where $P = diag\,(f, f, 1)[\,I_{3\times3} |\, 0\,]$
- Principal point offset: the origin of the camera coordinate system is not aligned with the ideal physical camera
- Thus (accounting for translation) $x_{3\times1} = K_{3\times3}\,[\,I_{3\times3}\,|\,0\,]X_{cam}$




$$K = \begin{bmatrix} f & & p_x \\ & f & p_y \\ & & 1 \end{bmatrix}$$




Where are we in this picture?

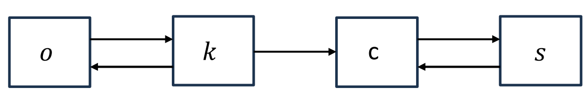
13

# CCD Camera Sensor Values

iphone 16 pr 460
Note 460ppi
5.89 → real
460 / inch
6.3
1+ 13 510
Sam S24 416
Moto 269

- $x_{3\times1} = k_{3\times3}\,[\,I_{3\times3}\,|\,0\,]X_{cam}$
- We now move to pixel units

$$K = \begin{bmatrix} m_x & & \\ & m_y & \\ & & 1 \end{bmatrix} \begin{bmatrix} f & & p_x \\ & f & p_y \\ & & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$




Where are we in this picture?

14

# World to Sensor

- World coordinate system
- Camera Coordinate System
  - Transformation written as a 3x3 matrix
  - Until this course:
    - Now we write a 4D vector in (homogeneous coordinates) related one coordinate to another in terms of 4x4 matrix relationship
    - Ĉ is the coordinate of the camera center in the world

$$X_{\text{cam}} = R(X - \hat{C})$$

$$X_{\text{cam}} = \begin{bmatrix} R & -R\hat{C} \\ 0 & 1 \end{bmatrix} X$$

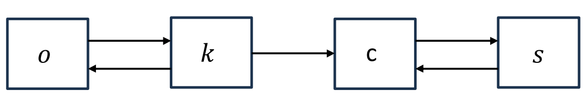
15

# World to Sensor

- World coordinate system
- Camera Coordinate System
  - Transformation written as a 3x3 matrix
  - Until this course:
    - Now we write a 4D vector in (homogeneous coordinates) related one coordinate to another in terms of 4x4 matrix relationship
    - Ĉ is the coordinate of the camera center in the world

$$X_{\text{cam}} = R(X - \hat{C})$$

$$X_{\text{cam}} = \begin{bmatrix} R & -R\hat{C} \\ 0 & 1 \end{bmatrix} X$$




Where are we in this picture?

16

# Camera Matrix Components

- We move to pixel units but further model a non-ideal sensor (a skew between the two axis, i.e., the two axis are not necessarily at right angles)
- Redefined $K$ is upper triangular
- homogenous $P = KR[I \mid -\hat{\mathrm{C}}] = K[R\ t]$

$$\begin{tikzcd} o \arrow[r, shift left] & k \arrow[l, shift left] \arrow[r] & c \arrow[r, shift left] & s \arrow[l, shift left] \end{tikzcd}$$

Where are we in this picture?

$$K = \begin{bmatrix} \alpha_x & s & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$

17

# Camera Matrix Components

- $x_{3\times1} = k_{3\times3}\ [\,I_{3\times3} \mid 0\,]X_{cam}$
- $x_{3\times1} = k_{3\times3}\ [\,I_{3\times3} \mid 0\,]X_{cam} = K_{3\times3}R\ [\,\mathrm{I}_{3\times3} \mid -\hat{\mathrm{C}}\,]X_w$
- $x = PX$
- $P = KR\big[\,I \mid -\hat{\mathrm{C}}\big|$ is the camera matrix (a finite projective camera)
- The set of camera matrices is identical with the set of homogenous $3 \times 4$ matrices for which the left $3 \times 3$ submatrix is non-singular
- $P$ can also be written as $P = K[R \mid t]$ where $t = -R\,\hat{\mathrm{C}}$

$$\begin{tikzcd} o \arrow[r, shift left] & k \arrow[l, shift left] \arrow[r] & c \arrow[r, shift left] & s \arrow[l, shift left] \end{tikzcd}$$

$$X_{\mathrm{cam}} = \begin{bmatrix} R & -R\tilde{C} \\ 0 & 1 \end{bmatrix} X_w$$

18